

Corrosion RC Beam Optimizer

Scientific Report - Full Pipeline

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This report presents the complete results of the Corrosion RC Beam Optimizer pipeline. The study applies ML ensemble models and symbolic regression (PySR) to predict the residual flexural capacity of corroded RC beams, benchmarked against ACI 318-19.

L1 target: $R2 \geq 0.9$ | L2 target: $R2 \geq 0.972$

Data: 804 specimens | Split: 70/30

1. ACI 318-19 Benchmark

R2: 0.8839

RMSE: 8.2289

MAE: 5.0617

MAPE: 29.9

ratio_mean: 1.1845

ratio_std: 0.5874

ratio_min: 0.2735

ratio_max: 5.2832

underestimate_pct: 48.5

n_specimens: 804

2. Ensemble Model Results (Part 1)

Best Model: CatBoost

Test R2 = 0.9757

Test RMSE = 3.9446 kN.m

Test MAE = 2.06 kN.m

Test CV% = 16.94%

Test SD/M = 0.1681

10-Fold CV (ALL 804 samples):

R2 = 0.9727

RMSE = 3.9891 kN.m

MAE = 1.9331 kN.m

CV% = 17.57%

SD/M = 0.1753

3. PySR Symbolic Regression (Part 2)

Winner: DIRECT

Equation: $1.44054648571482 \cdot (d + f_c - 170.12386) \cdot (-0.0035611875 \cdot \sqrt{\text{CSI}}) + \text{RI} + 0.0012547432 \cdot b - 1/f$

Mmax R2 = 0.9351

Mmax RMSE = 6.1530 kN.m

Mmax MAE = 4.0830 kN.m

Mmax MAPE = 35.62%

CV% = 27.11%

SD/M = 0.2711

L1 broken = True

L2 broken = False

Ratio approach R2 = 0.9272

Direct approach R2 = 0.9351

4. Pareto Front Equations

C= 1 | Mmax R2=0.0708 | MAPE=89.0%

C= 3 | Mmax R2=0.5040 | MAPE=185.4%

C= 4 | Mmax R2=0.5162 | MAPE=165.7%

C= 5 | Mmax R2=0.7893 | MAPE=50.3%

C= 6 | Mmax R2=0.8614 | MAPE=39.6%

C= 7 | Mmax R2=0.8620 | MAPE=42.8%

C= 8 | Mmax R2=0.8803 | MAPE=40.9%

C= 9 | Mmax R2=0.8897 | MAPE=35.3%

C=11 | Mmax R2=0.9048 | MAPE=36.8%

C=13 | Mmax R2=0.9175 | MAPE=35.1%

C=15 | Mmax R2=0.9261 | MAPE=35.0%

C=16 | Mmax R2=0.9273 | MAPE=31.9%

C=18 | Mmax R2=0.9287 | MAPE=31.3%

C=19 | Mmax R2=0.9320 | MAPE=31.1%

C=21 | Mmax R2=0.9336 | MAPE=29.1%

C=23 | Mmax R2=0.9348 | MAPE=35.9%

C=24 | Mmax R2=0.9351 | MAPE=35.6% <<<

C=25 | Mmax R2=0.9351 | MAPE=34.7%

5. Figures Gallery

Fig10 Pareto Mmax Performance

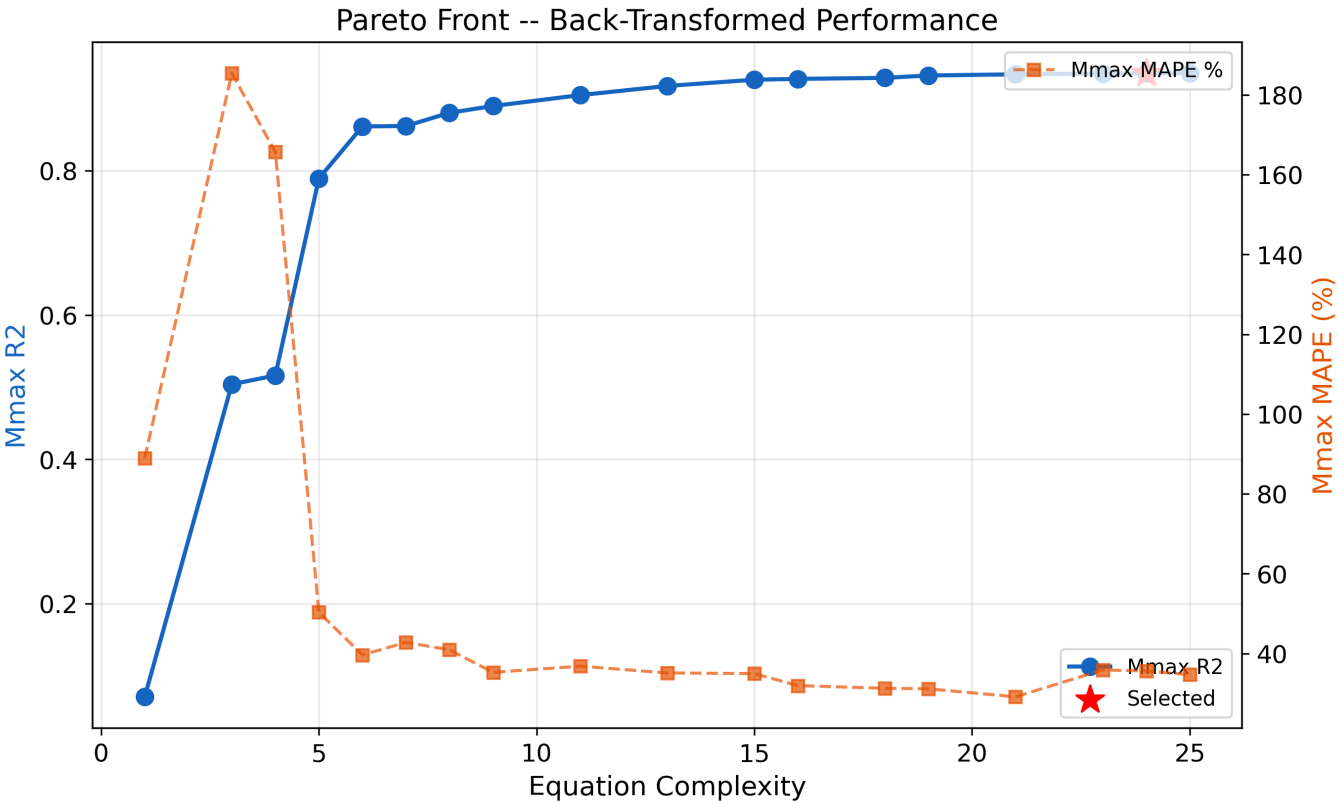


Fig11 Ratio Vs Eta

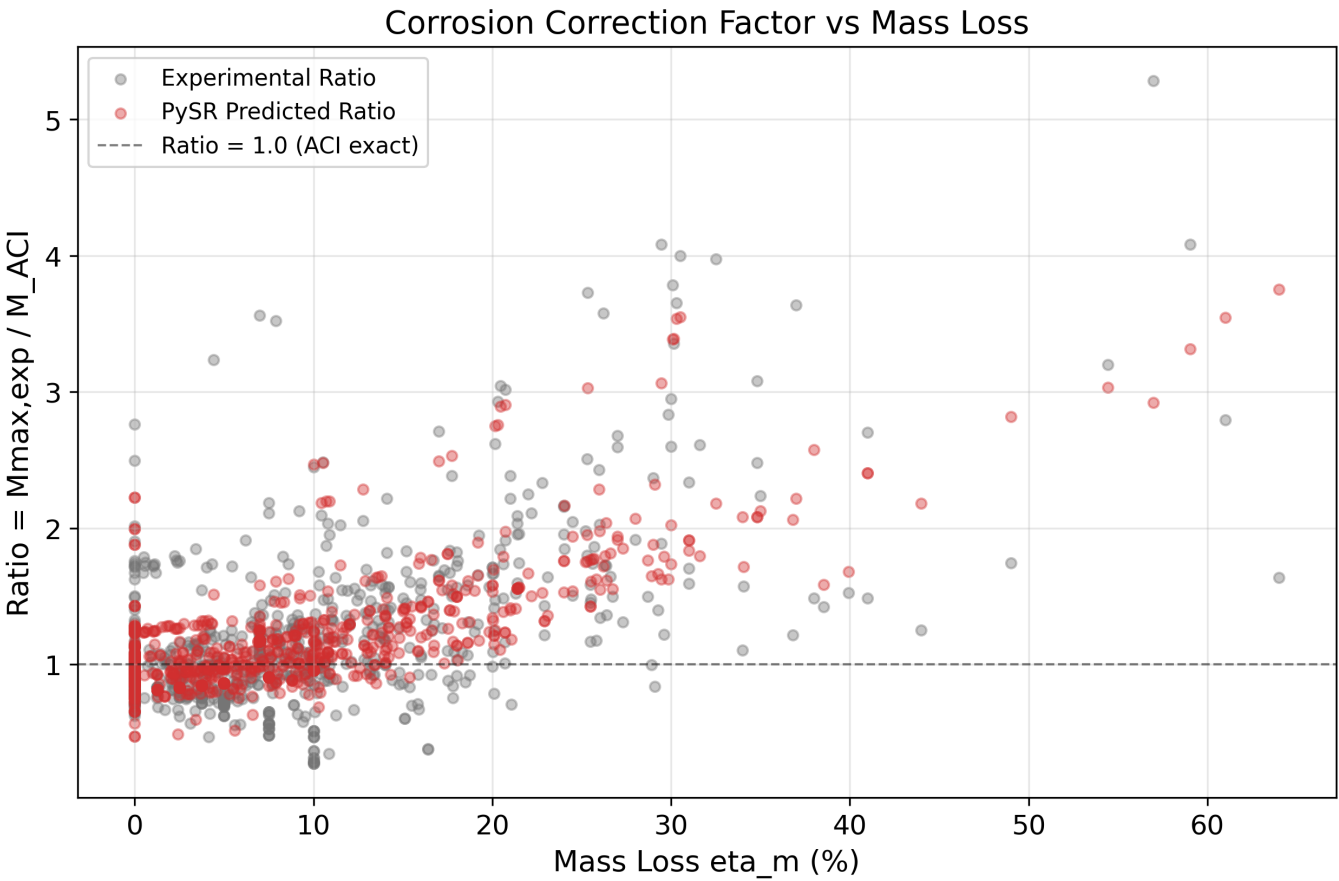


Fig1 Predicted Vs Experimental

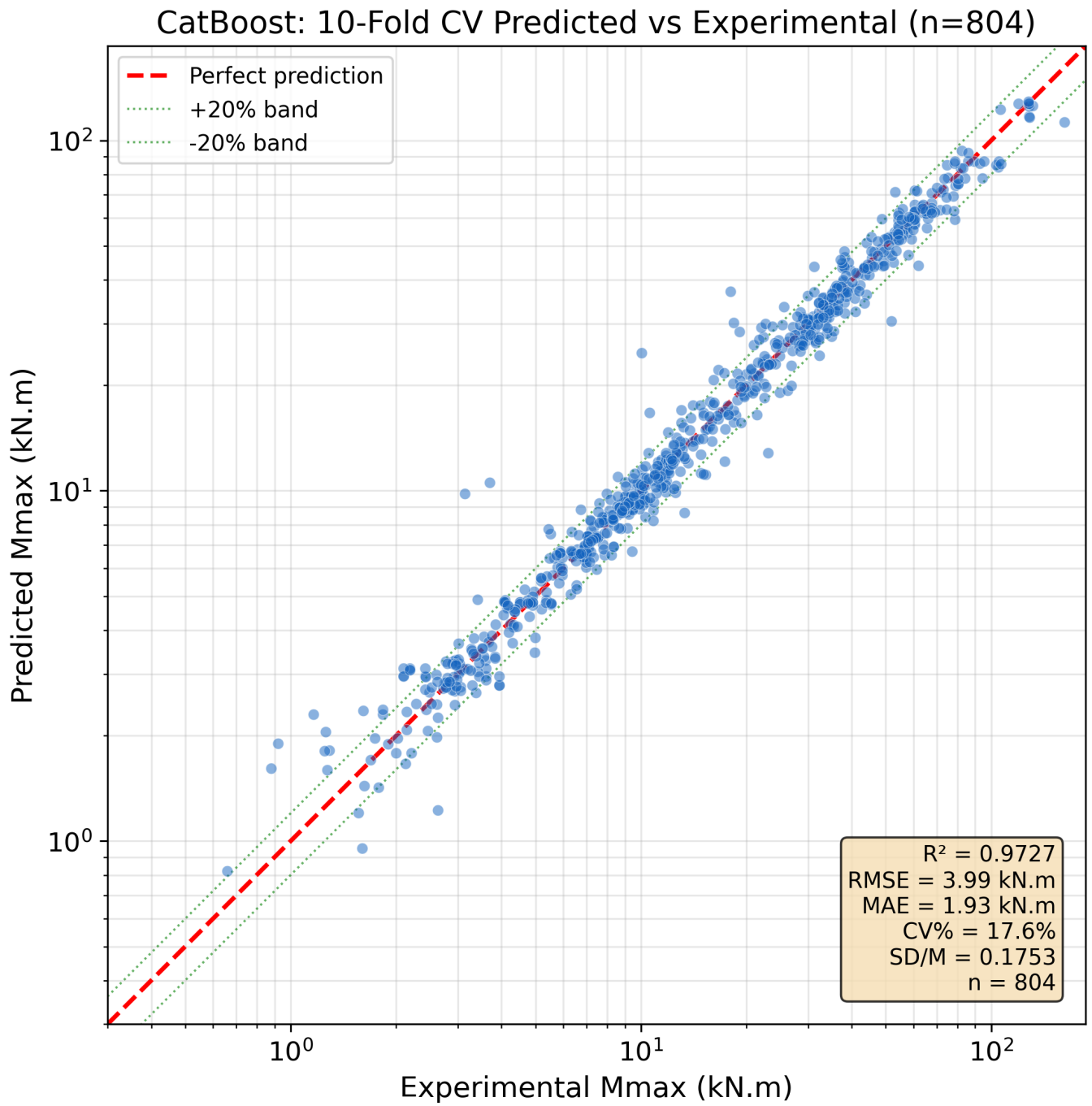


Fig1B Linear Scatter

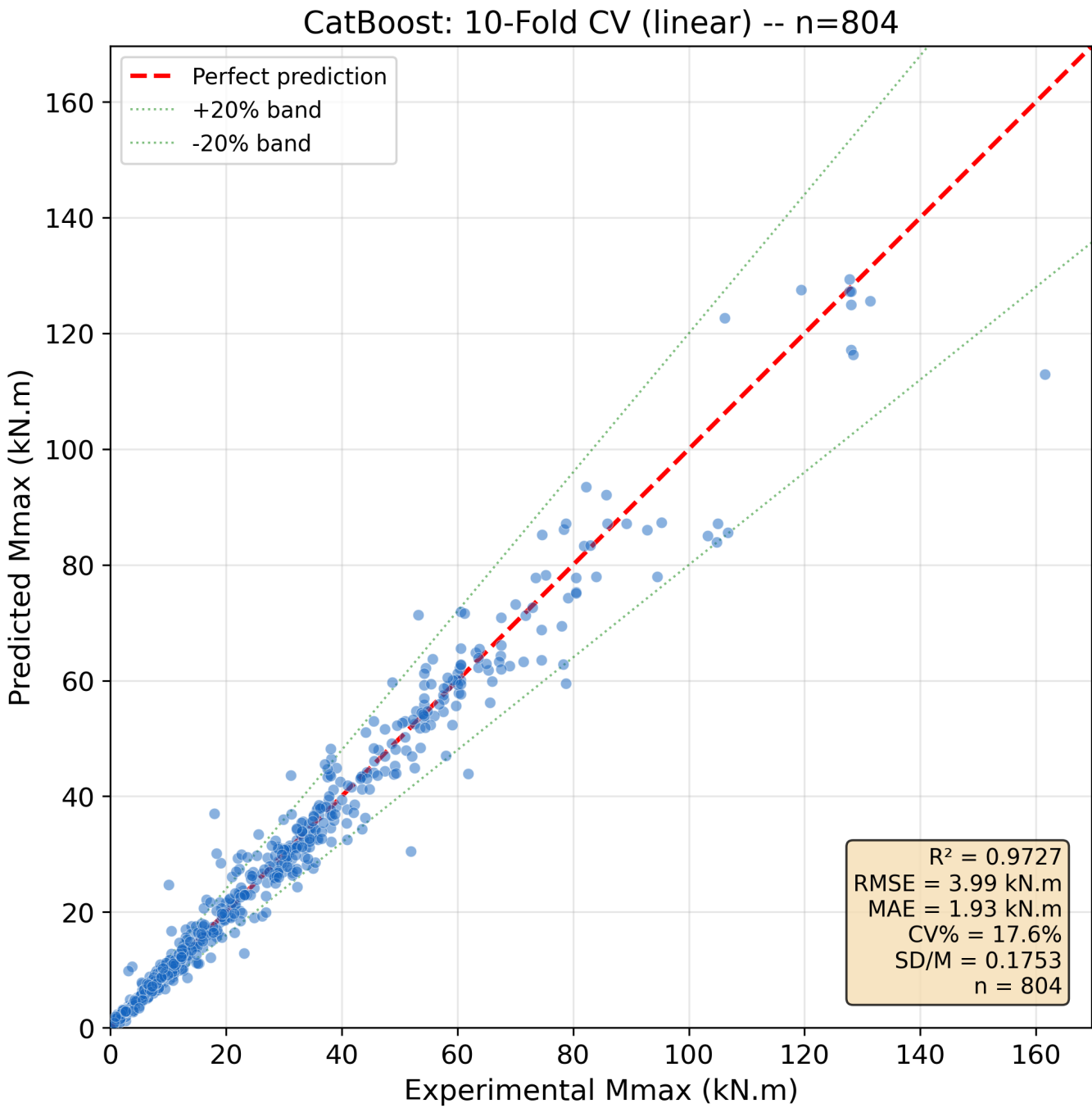


Fig2 Test Set Scatter

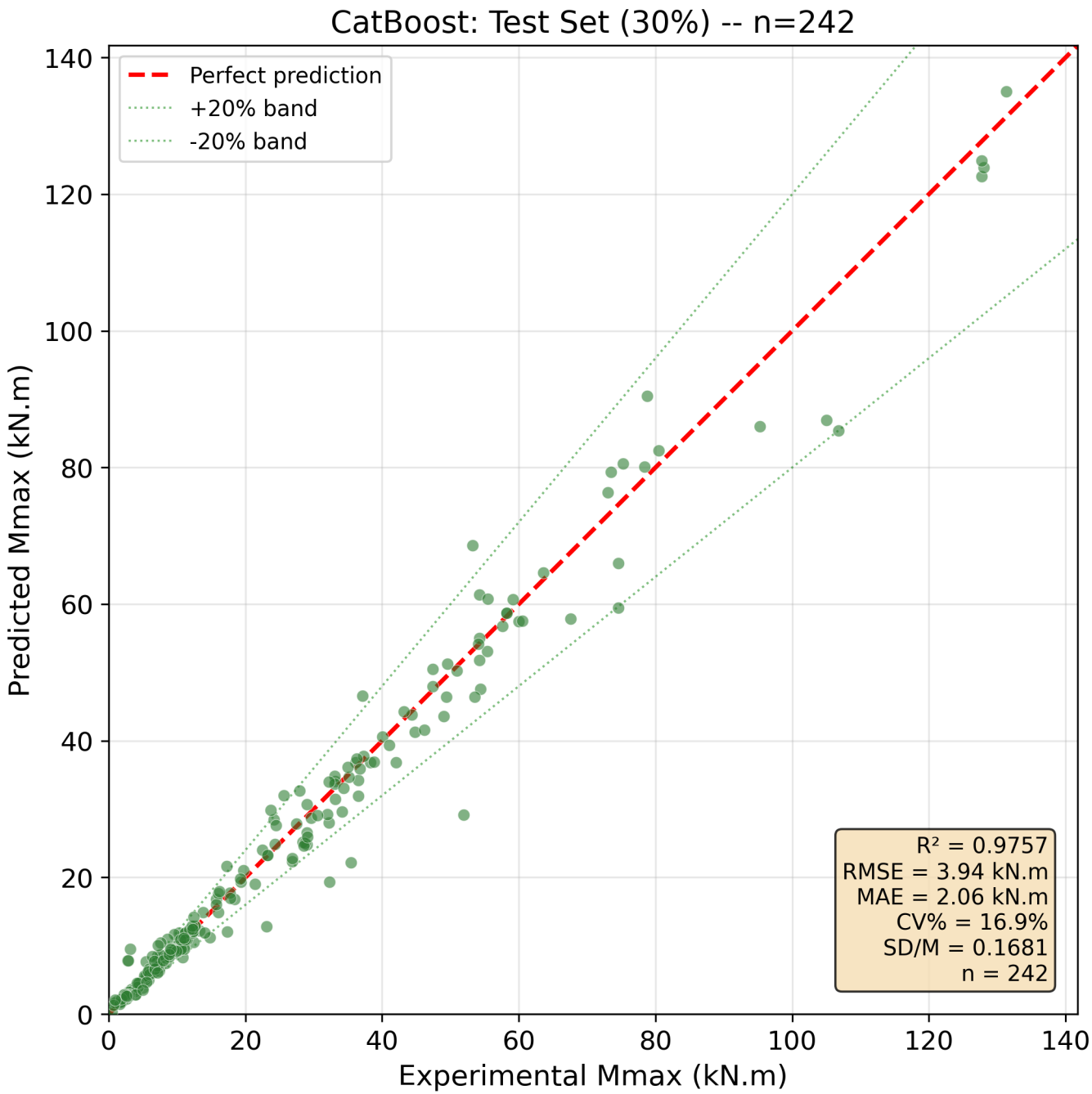


Fig3 Ensemble Vs Aci Scatter

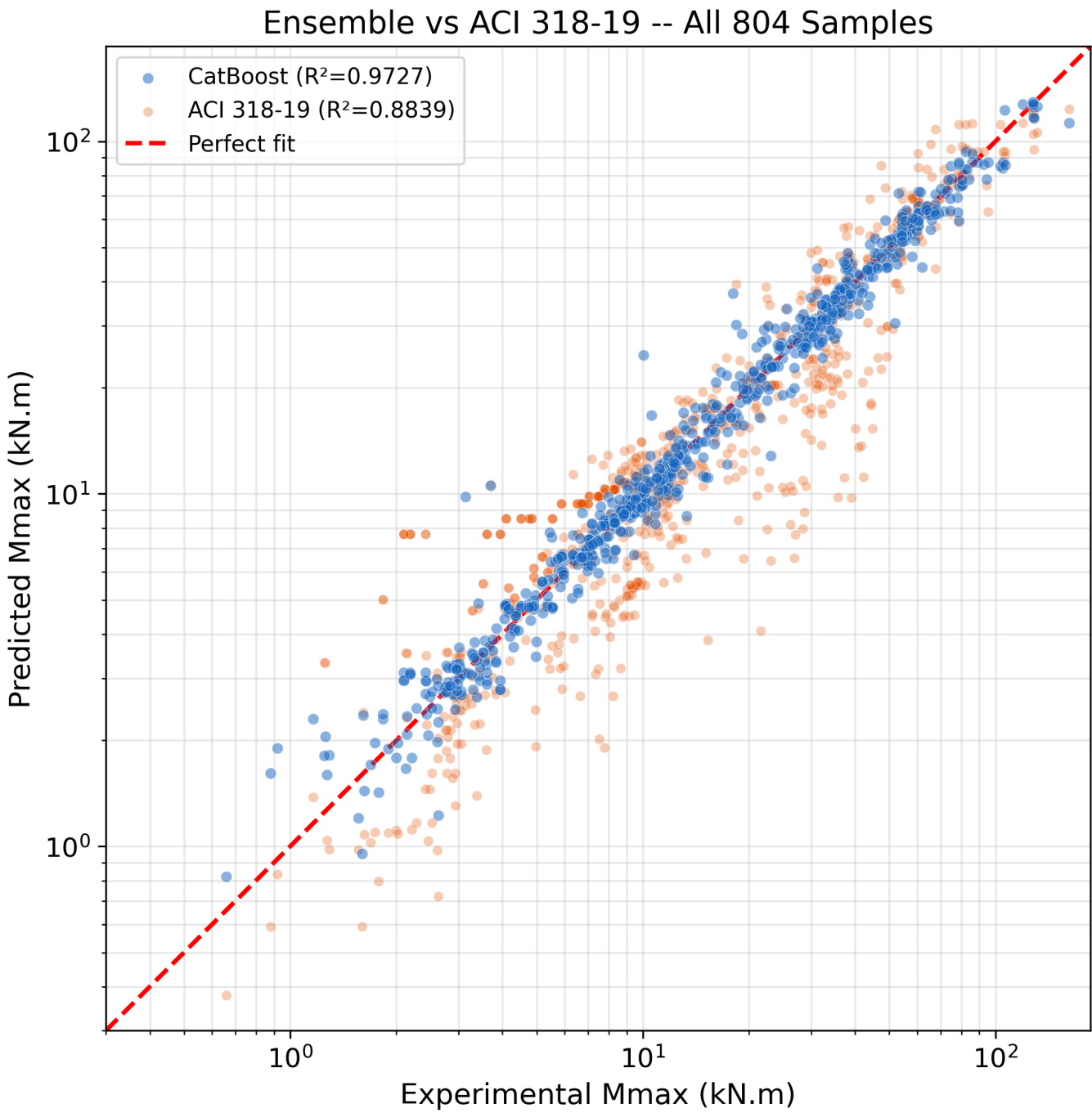


Fig4 Kfold Boxplot

10-Fold CV: $R^2 = 0.9723 \pm 0.0153$

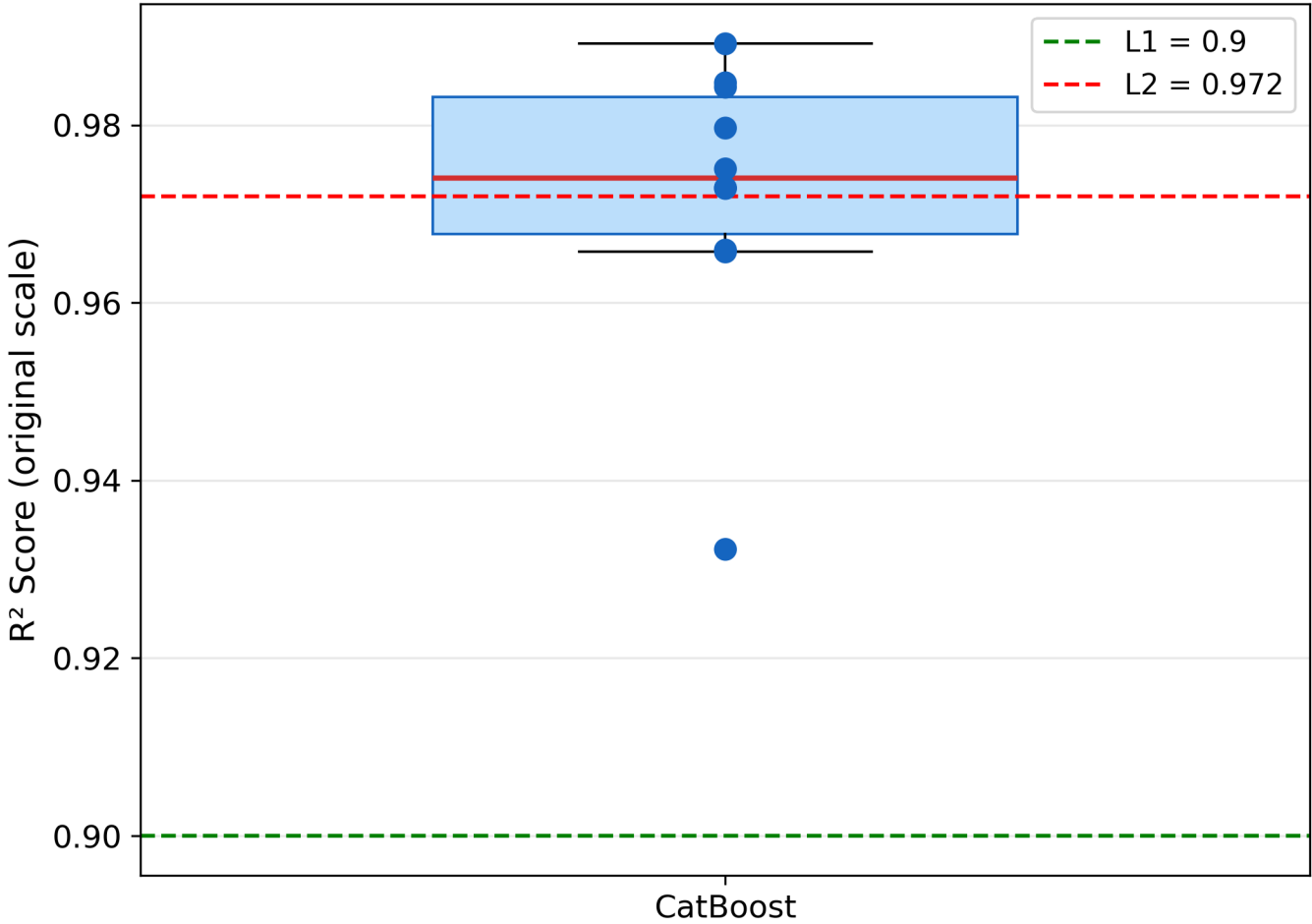


Fig5 Error Distribution

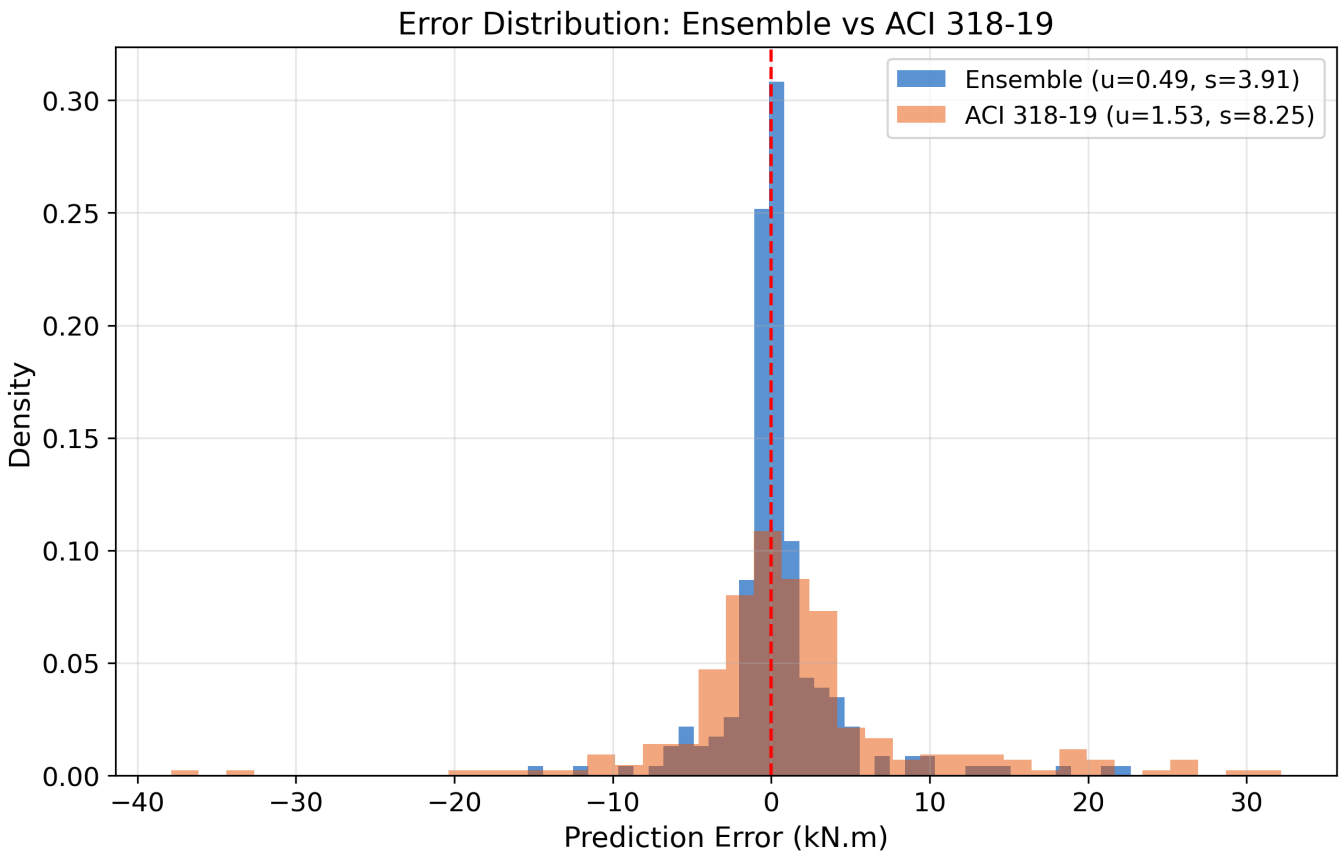


Fig6 Model Comparison

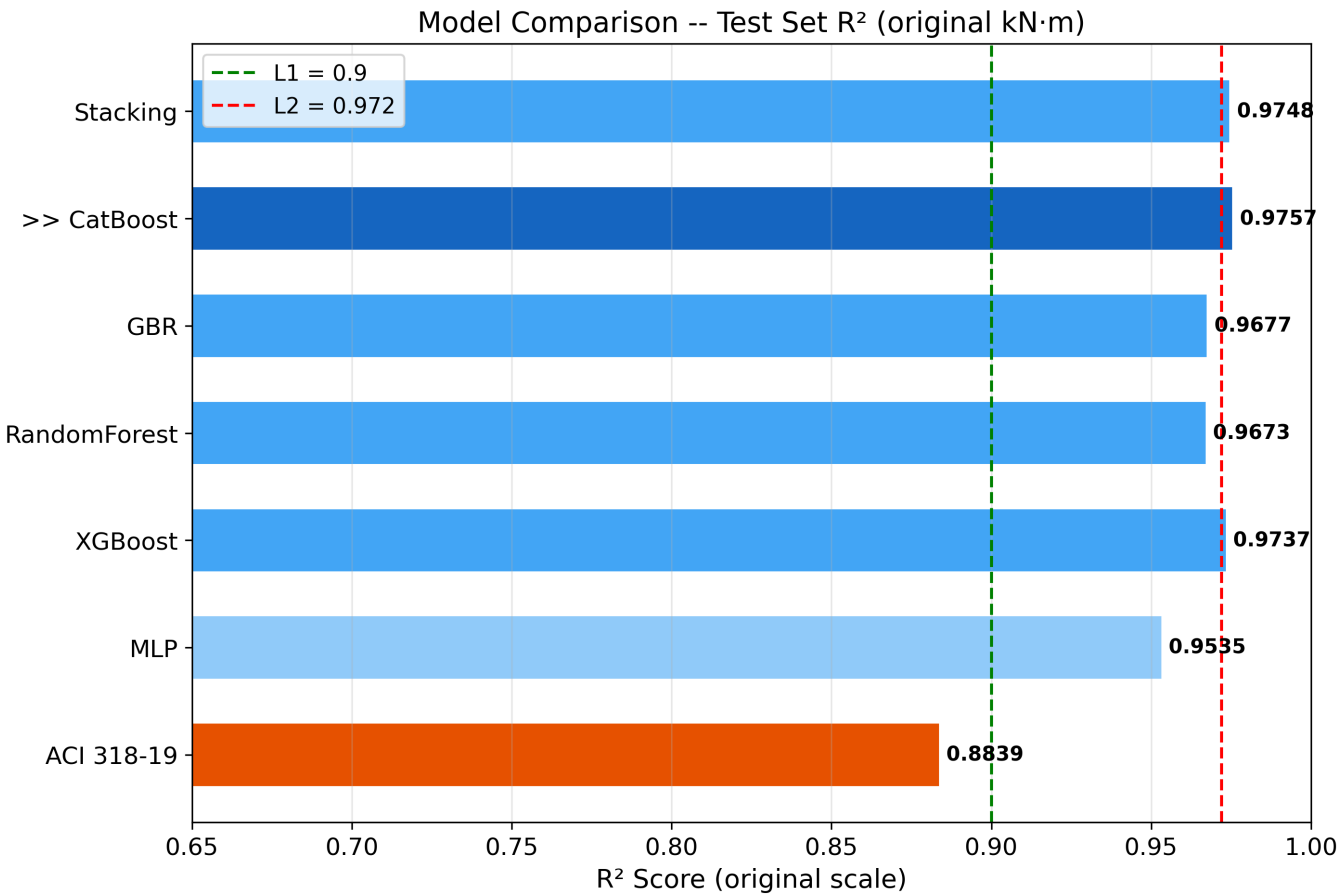


Fig7 Taylor Diagram

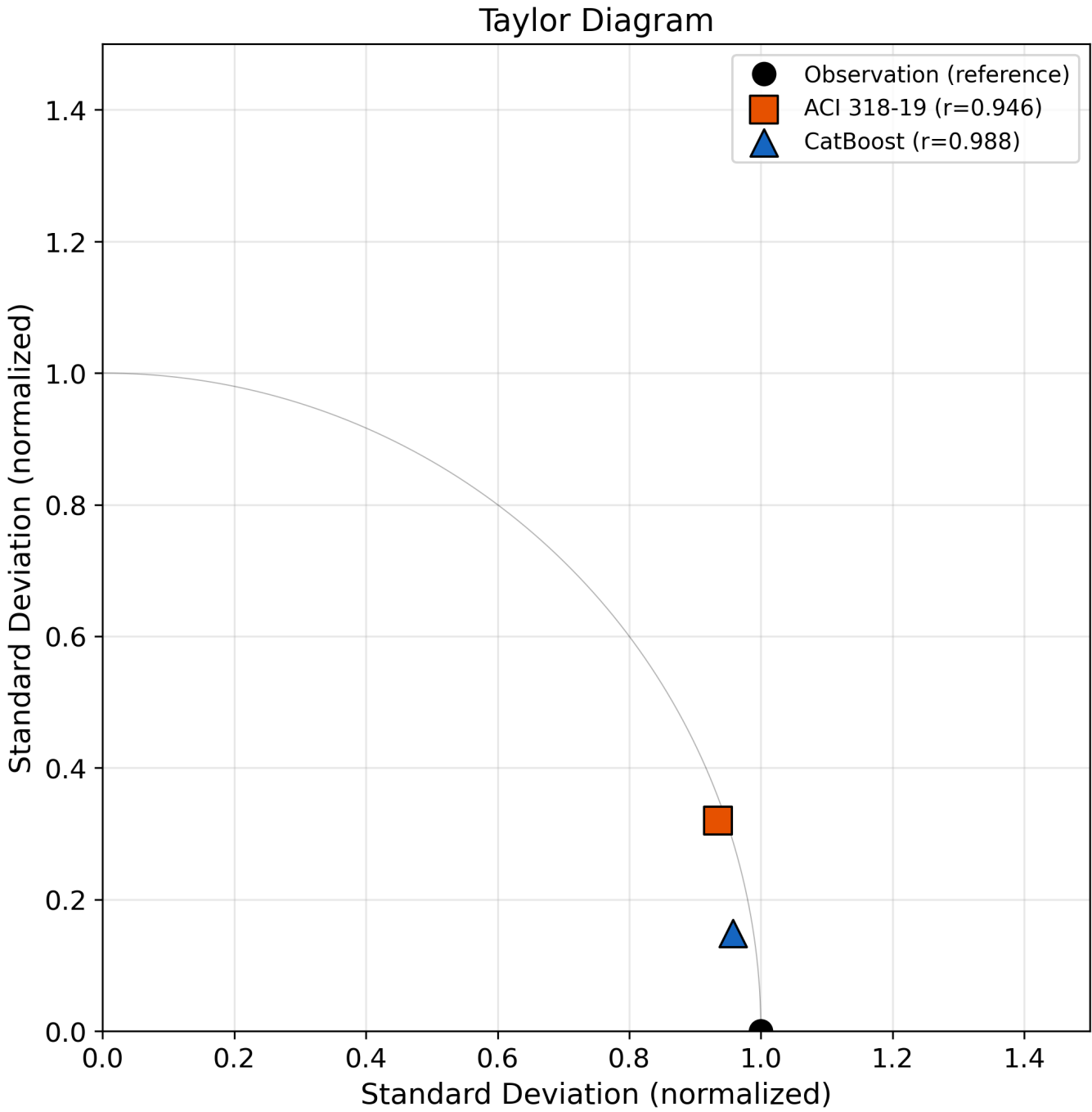


Fig8 Pysr Equation Scatter

PySR DIRECT Equation: Predicted vs Experimental
 $R^2=0.9351$ | $RMSE=6.15$ | $MAPE=35.6\%$ | $n=804$

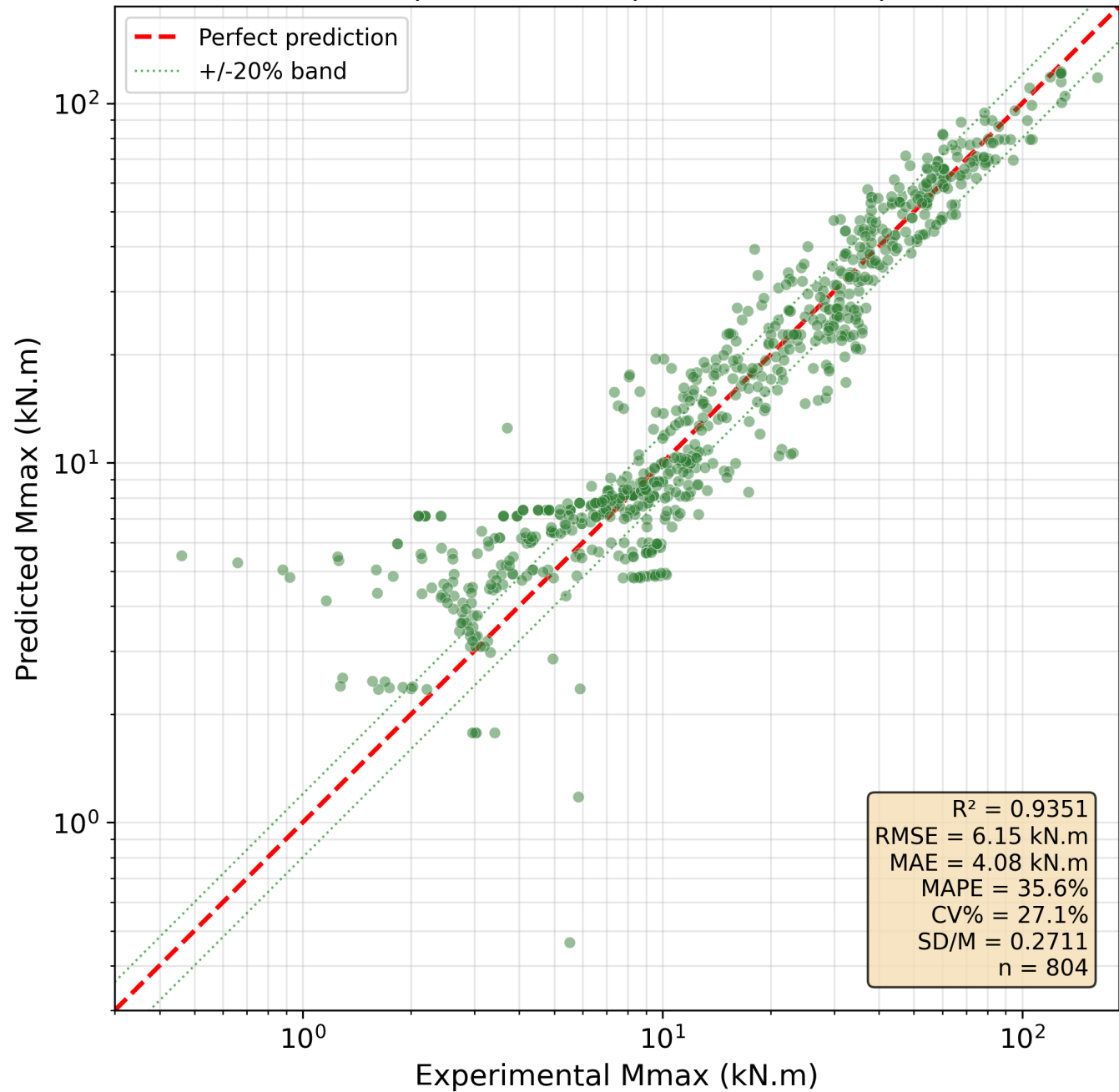
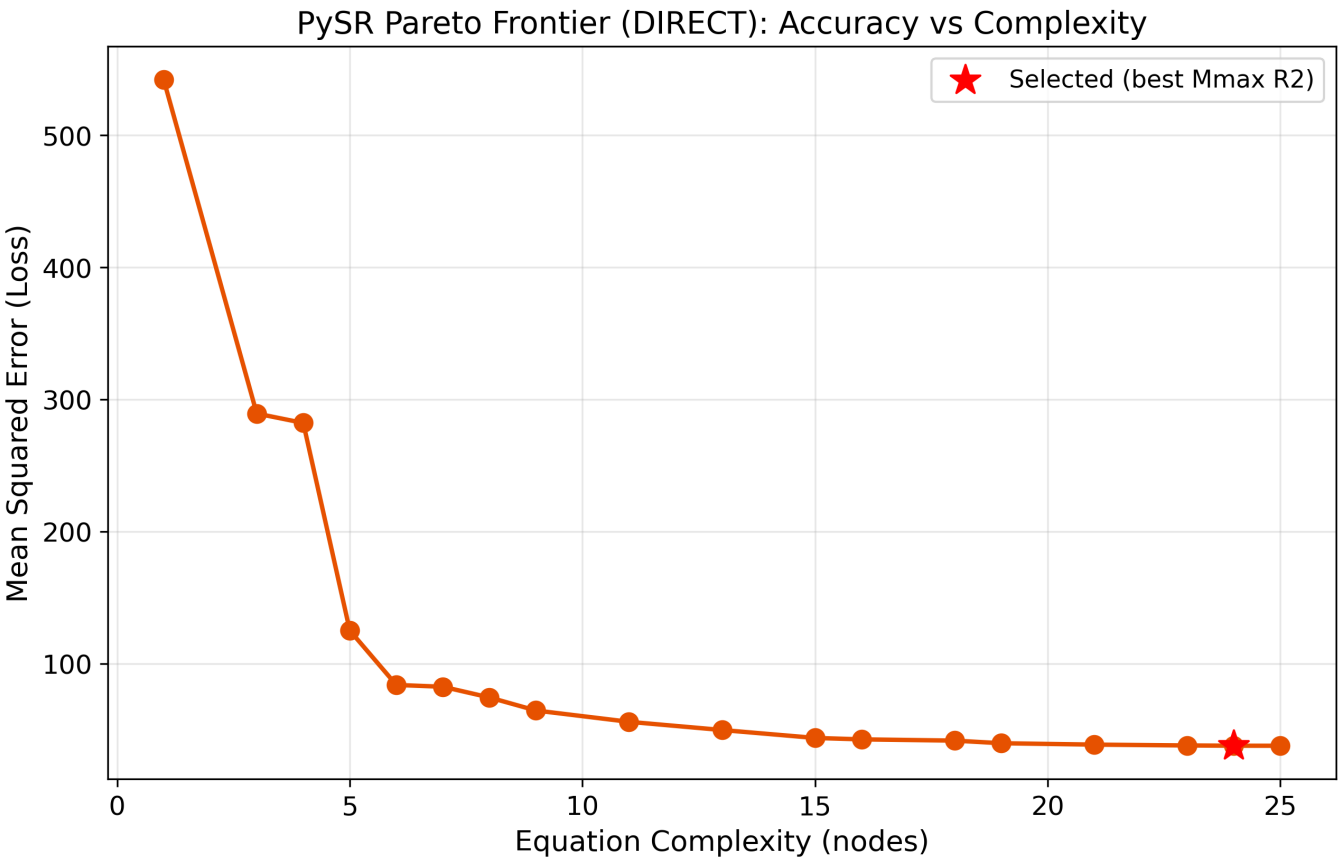


Fig9 Pareto Curve



Shap Beeswarm

SHAP Beeswarm Plot — Feature Impact on R(%) Prediction

